



Experiment 4

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Subject Name: Full Stack Development-II

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1. Aim:

To optimize the performance of the EcoTrack React application using memoization techniques and code splitting, and to enhance the user interface using enterprise-grade Material UI components.

2. Objective:

- Understand the causes of unnecessary re-renders in React applications
- Optimize React components using React.memo to prevent avoidable re-renders
- Apply useMemo to efficiently compute derived data and avoid redundant calculations
- Use useCallback to memoize event handler functions and improve component performance
- Implement lazy loading of components and routes using React.lazy and Suspense
- Reduce initial bundle size and improve application load performance through code splitting
- Enhance the visual appearance and usability of the EcoTrack application using material UI components
- Design a clean, consistent, and responsive user interface using Material UI layouts
- and typography

3. Implementation/Code:

login.jsx:

```
import { useAuth } from "../context/AuthContext";  
import { useNavigate } from "react-router-dom";  
import Button from '@mui/material/Button';
```

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```
import Box from '@mui/material/Box';
import Typography from '@mui/material/Typography';
const Login = () => {
  const { login } = useAuth();
  const navigate = useNavigate();
  const handleLogin = () => {
    login();
    navigate("/dashboard");
  };
  return (
    <Box sx={{
      display: 'flex',
      flexDirection: 'column',
      alignItems: 'center',
      justifyContent: 'center',
      minHeight: '100vh',
      p: 3
    }}>
      <Typography variant="h4" gutterBottom>
        Welcome to EcoTrack
      </Typography>
      <Button
        variant="contained"
        color="primary"
        size="large"
        onClick={handleLogin}
        sx={{ mt: 3, px: 6, py: 1.5 }}
      >
        Login to EcoTrack
      </Button>
    </Box>
  );
}
```

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```
);  
};  
export default Login;
```

dashboard.jsx:

```
import { useState } from 'react';  
import { Outlet, useNavigate, Link as RouterLink } from 'react-router-dom';  
import { useAuth } from "../context/AuthContext";  
import Header from "../components/Header";  
import AppBar from '@mui/material/AppBar';  
import Box from '@mui/material/Box';  
import Toolbar from '@mui/material/Toolbar';  
import Typography from '@mui/material/Typography';  
import Drawer from '@mui/material/Drawer';  
import List from '@mui/material/List';  
import ListItem from '@mui/material/ListItem';  
import ListItemButton from '@mui/material/ListItemButton';  
import ListItemIcon from '@mui/material/ListItemIcon';  
import ListItemText from '@mui/material/ListItemText';  
import Divider from '@mui/material/Divider';  
import IconButton from '@mui/material/IconButton';  
import Button from '@mui/material/Button';  
import HomeIcon from '@mui/icons-material/Home';  
import DashboardIcon from '@mui/icons-material/Dashboard';  
import AssessmentIcon from '@mui/icons-material/Assessment';  
import LogoutIcon from '@mui/icons-material/Logout';  
import MenuIcon from '@mui/icons-material/Menu';  
const drawerWidth = 240;  
const Dashboard = () => {  
  const { logout } = useAuth();  
  const navigate = useNavigate();  
  const [mobileOpen, setMobileOpen] = useState(false);  
  const handleLogout = () => {  
    logout();  
    navigate("/login");  
  };  
};
```

```
const handleDrawerToggle = () => {
  setMobileOpen(!mobileOpen);
};
const drawerContent = (
  <div>
    <Toolbar /> { /* spacer under AppBar */}
    <Divider />
    <List>
      <ListItem disablePadding>
        <ListItemButton
          component={RouterLink}
          to=""
          onClick={() => setMobileOpen(false)}
        >
          <ListItemIcon><HomeIcon /></ListItemIcon>
          <ListItemText primary="Home" />
        </ListItemButton>
      </ListItem>
      <ListItem disablePadding>
        <ListItemButton
          component={RouterLink}
          to="overview"
          onClick={() => setMobileOpen(false)}
        >
          <ListItemIcon><DashboardIcon /></ListItemIcon>
          <ListItemText primary="Overview" />
        </ListItemButton>
      </ListItem>

      <ListItem disablePadding>
        <ListItemButton
          component={RouterLink}
          to="reports"
          onClick={() => setMobileOpen(false)}
        >
          <ListItemIcon><AssessmentIcon /></ListItemIcon>
          <ListItemText primary="Reports" />
        </ListItemButton>
      </ListItem>
    </List>
  </div>
);
```



```
<Divider />
<List>
  <ListItem disablePadding>
    <ListItemButton onClick={handleLogout} sx={{ color: 'error.main' }}>
      <ListItemIcon><LogoutIcon color="error" /></ListItemIcon>
      <ListItemText primary="Logout" />
    </ListItemButton>
  </ListItem>
</List>
</div>
);

return (
  <Box sx={{ display: 'flex' }}>
    { /* AppBar at top */ }
    <AppBar
      position="fixed"
      sx={{
        width: { sm: `calc(100% - ${drawerWidth}px)` },
        ml: { sm: `${drawerWidth}px` }, // ← important: push content right
        backgroundColor: '#4CAF50',
      }}
    >
      <Toolbar>
        <IconButton
          color="inherit"
          edge="start"
          onClick={handleDrawerToggle}
          sx={{ mr: 2, display: { sm: 'none' } }}
        >
          <MenuIcon />
        </IconButton>
        <Typography variant="h6" noWrap component="div">
          Dashboard
        </Typography>
      </Toolbar>
    </AppBar>

    { /* Sidebar - LEFT side */ }
    <Box
      component="nav"
      sx={{
        width: { sm: drawerWidth },

```



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```
flexShrink: { sm: 0 }
}}
>
{/* Mobile temporary drawer */}
<Drawer
variant="temporary"
open={mobileOpen}
onClose={handleDrawerToggle}
ModalProps={{ keepMounted: true }}
sx={{
display: { xs: 'block', sm: 'none' },
'& .MuiDrawer-paper': {
boxSizing: 'border-box',
width: drawerWidth,
backgroundColor: '#4CAF50', // green theme
color: 'white',
},
}}
>
{drawerContent}
</Drawer>
{/* Desktop permanent drawer - LEFT */}
<Drawer
variant="permanent"
sx={{
display: { xs: 'none', sm: 'block' },
'& .MuiDrawer-paper': {
boxSizing: 'border-box',
width: drawerWidth,
backgroundColor: '#4CAF50',
color: 'white',
},
}}
Open
>
{drawerContent}
</Drawer>
</Box>
{/* Main content - pushed to right */}
<Box
component="main"
sx={{
flexGrow: 1,
```

```
p: 3,  
width: { sm: `calc(100% - ${drawerWidth}px)` },  
ml: { sm: `${drawerWidth}px` }, // ← ensures shift  
mt: '64px', // space under AppBar (Toolbar height)  
backgroundColor: 'background.default',  
}}  
>  
{/* Spacer for fixed AppBar */}  
<Toolbar />  
{/* This is where your child routes render: DashboardHome / Overview / Reports  
*/}  
<Outlet />  
</Box>  
</Box>  
);  
};  
export default Dashboard;
```

App.jsx:

```
import { lazy, Suspense } from "react";  
import { Routes, Route, Navigate } from "react-router-dom";  
import ProtectedRoute from "../routes/ProtectedRoute";  
// Lazy load pages  
const Dashboard = lazy(() => import("../pages/Dashboard"));  
const DashboardHome = lazy(() => import("../pages/DashboardHome"));  
const Overview = lazy(() => import("../pages/Overview"));  
const Reports = lazy(() => import("../pages/Reports"));  
const Login = lazy(() => import("../pages/Login"));  
const App = () => {  
  return (  
    <Suspense fallback={  
      <div style={{  
        display: 'flex',  
        justifyContent: 'center',  
        alignItems: 'center',  
        minHeight: '100vh'  
      }}>  
        Loading application...  
      </div>  
    }>  
    <Routes>
```

```
<Route path="/login" element={<Login />} />
<Route
  path="/dashboard"
  element={
    <ProtectedRoute>
      <Dashboard />
    </ProtectedRoute>
  }
/>
<Route index element={<DashboardHome />} />
<Route path="overview" element={<Overview />} />
<Route path="reports" element={<Reports />} />
</Route>
<Route path="*" element={<Navigate to="/dashboard" />} />
</Routes>
</Suspense>
);
};
export default App;
```

Main.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import { BrowserRouter } from "react-router-dom";
import { AuthProvider } from "./context/AuthContext";
import { Provider } from "react-redux";
import { store } from "./redux/store";
import { ThemeProvider } from '@mui/material/styles';
import CssBaseline from '@mui/material/CssBaseline';
import theme from "./theme";
import "./index.css";
import "@fontsource/roboto/300.css";
import "@fontsource/roboto/400.css";
import "@fontsource/roboto/500.css";
import "@fontsource/roboto/700.css";
ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <Provider store={store}>
      <ThemeProvider theme={theme}>
```




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```
<CssBaseline />  
<BrowserRouter>  
<AuthProvider>  
<App />  
</AuthProvider>  
</BrowserRouter>  
</ThemeProvider>  
</Provider>  
</React.StrictMode>  
);
```

4. Output:

Overview Page

This is the Overview section of EcoTrack.

Reports & Analytics

Detailed reports and carbon footprint analysis coming soon...



5. Learning Outcome:

- Identify and analyze causes of unnecessary re-renders in a React application.
- Apply React.memo, useMemo, and useCallback to optimize component rendering performance.
- Implement code splitting and lazy loading using React.lazy and Suspense to improve load time.
- Design a responsive and consistent UI using Material UI components and theming.
- Integrate performance optimization techniques with secure routing and modern React architecture.